

Sub-tropical High Pressure Belt

- It extends between 30° to 35° N and S in both the hemispheres. The high pressure along this belt is due to subsidence of air coming from equatorial regions and also due to blocking effect of coriolis force.
- The convergence of winds at higher altitude above this zone results in the subsidence of air from higher altitudes. Thus, descent of wind results in concentration of their volume and ultimately causes high pressure. This zone of high pressure is also called as Horse Latitude.

Sub Polar Low Pressure Belt

- It extends between 60° to 65° in both the hemisphere. The low pressure belt does not appear to be thermally induced because there is low temperature throughout the year and as such there should have been high pressure belt instead of low pressure belt. Thus, it is dynamically induced.
- It is more developed and regular in Southern hemisphere than in Northern hemisphere because of the over dominance of water (ocean) in the Southern hemisphere.

Polar High

High pressure persists at the poles throughout the year because of the prevalence of very low temperature all the year round. Polar highs are small in area and extend around the poles.

Shifting of Pressure Belts

- The entire system of pressure and wind belts follows the movement of mid-day Sun. In June, when the overhead Sun is over the Tropic of Cancer, all the belts move about 5°–10° North of their average position.
- In the same manner, when the Sun is overhead at the Tropic of Capricorn in December, all the belts swings 5°–10° South of their average position.

Wind System

The pressure difference is the major cause of the genesis of the wind system. The air moves from high pressure to low pressure.

- The slope of pressure from high to low is called as Pressure Gradient, which is also called as Barometric Slope.
- The imaginary line joining the points having same pressure is called Isobars. The direction of air movement should be perpendicular to the isobars because the direction of pressure gradient is perpendicular to the isobars, but the direction is deviated from the expected one due to **coriolis force** caused by the rotation of the Earth.

Wind Direction and Related Laws

- The Coriolis force generated due to the rotation of Earth acts as a deflective force to the wind direction.
- Because of the coriolis force, all the winds are deflected to the right in the Northern hemisphere while they are deflected to the left in the Southern hemisphere with respect to the rotating Earth. This is referred to as **Ferrel's law**.
- The coriolis force is absent along the equator, but increases progressively towards the poles.

Types of Winds

(i) Permanent Winds

These winds include trade wind, westerlies and polar winds, these are as follows :

- **Trade Wind** These are steady currents of air blowing from the sub-tropical high pressure belt towards the equatorial low pressure belt. Under the influence of the coriolis forces they flow from the North-East in the Northern hemisphere and from South-East in the Southern hemisphere.
- **Westerlies** The permanent winds blowing from the sub-tropical high pressure belt to the sub polar low pressure belt in both the hemisphere is called Westerlies. The general direction of the Westerlies is South-West to North-East in the Northern hemisphere and North-West to South-East in the Southern hemisphere. Westerlies are best developed between 40° and 65° S latitude, these latitudes are often called Roaring forties, furious fifties and shrieking sixties.
- **Polar Winds** These winds blow from polar high pressure belt to sub polar low pressure belt. They are North Easterly in Northern hemisphere and South Easterly in the Southern hemisphere.

(ii) Seasonal Winds

The winds, which reverse its direction completely every 6 months are called Seasonal Winds. The best example is monsoon winds.

(iii) Local Winds

Local winds are horizontal movement of winds on small-scale. Movement of these winds is mainly caused by local factors such as water body, mountain etc.

Winds	Nature	Region
Land breeze	Warm	Experienced along coastal areas. Land breeze blows from land to sea.
Sea breeze	Cold	Experienced along coastal areas. Sea breeze blows from sea to land.
Chinook (snow eater)	Warm	Rockies (USA and Canada)
Sirocco	Hot	North Africa
Fohn	Warm	Alps/Europe
Khamsin	Hot	Egypt
Blizzard	Cold	Siberian, Canada, USA
Bora	Cold	Yugoslavia
Southerly burster	Cold	Australia
Purga	Cold	Russian, Tundra
Bire	Cold	France
Cape doctor	Cold	South Africans's coast
Harmattan (the doctor)	Hot	Sahara to Guinea coast (Ghana, Nigeria, etc)
Zonda	Warm	Argentina, Chile/Andes
Brick fielder	Hot	Australia
Samun	Hot	Iran
Levanter	Cold Wind	Spain
Norwester	Hot Wind	New Zealand
Leveche	Hot	Algeria, Morocco
Santa Ana	Warm	USA (California)
Berg	Warm	South Africa
Yoma	Warm	Japan
Karaburan	Hot	Tarim basin
Black roller	Hot/Dusty	North America
Kalbaisakhi/ Bardoli Chheerha	Hot	North India
Mistral	Cold	France to Mediterranean (Rhine valley)
Loo	Hot/Dry	North Indian Plains (Punjab to Bihar)

Jet Stream

- The strong and rapidly moving circumpolar Westerly air circulation in a narrow belt of a few 100 km width in the upper limit of troposphere is called Jet Stream. Their circulation path is wavy and meandering.
- The extent of the jet streams narrows down during the summer season because of their Northward shifting while these extend up to 20° North latitude during winter season.

Cyclones

- Cyclones are the centres of low pressure surrounded by closed isobars having increasing pressure outward and closed air circulation from outside towards the central low pressure in such a way that air blows inward direction.
- Air blows inward in anti-clockwise direction in the Northern hemisphere and clockwise direction in the Southern hemisphere.

Types of Cyclone

Cyclone are mainly of two types :

- Tropical Cyclone** They have thermal origin, they acquire whirling motion due to coriolis force. Wind speed varies from zero to 1200 km/h
- Temperate Cyclone** They develop over mid-latitudes. Range 35° to 65° N and S. They are also called extra tropical or wave cyclones.

Cyclone	Region
Typhoons	China sea
Tropical cyclones	Indian ocean
Hurricanes	Caribbean sea
Tornadoes	USA
Willy Willies	Northern Australia

Anticyclone

- Anticyclones are the wind system, which have the highest air pressure at the centre and lowest at the outer margins surrounded by circular isobars.
- Air blows from centre to outward in clockwise direction in Northern hemisphere and anti-clockwise direction in Southern hemisphere.
- They are high pressure system and common in sub-tropical belts and practically absent in the equator region. They are generally associated with rainless fair weather and that's why they are called as Weatherless Phenomena.

Thunderstorms

- Thunderstorms are local storms characterised by swift upward movement of air and heavy rainfall with cloud thunder and lighting.
- Structurally, thunderstorms consist of several convective cells, which are characterised by strong updraft of air.

Tornado

- Tornadoes are very strong tropical cyclones of smaller size. In the Mississippi valley (US), they are called Twisters.
- They are more destructive than cyclones as the speed of winds is very high, exceeding 320 km per hour.

Humidity and Related Aspects

- Humidity of air refers to the content of the water vapour present in the air at a particular time and place.
 - Humidity is measured by an instrument called Hygrometer. Another instrument used for the same purpose is **sling psychrometer**.
 - *The atmospheric humidity is expressed in a number of ways such as :*
 - **Humidity Capacity** The capacity of air of certain volume at certain temperature to retain maximum amount of moisture content.
 - **Absolute Humidity** The total weight of moisture content per volume of air at definite temperature is called Absolute Humidity.
 - **Specific Humidity** The mass of the water vapour in grams contained in a kilogram of air and it represents the actual quantity of moisture present in a definite air.
 - **Relative Humidity** It is the ratio of the amount of water vapour actually present in the air having definite volume and temperature (i.e. absolute humidity) to the maximum amount the air can hold (i.e. humidity capacity).
- $$\text{Relative Humidity} = \frac{\text{Absolute Humidity}}{\text{Humidity Capacity}} \times 100$$
- **Condensation** is the change of physical state of matter from gaseous phase into liquid phase. It is the reverse of vaporisation.
 - When the relative humidity reaches 100%, the air is completely saturated. The air temperature is said to be as dew point.

- **Fog** is made from the droplets of water suspended in the lower layer of the atmosphere. Fog is not considered as a form of precipitation. Visibility of less than 1 km is the internationally recognised definition of fog.
- **Smog** (Smoke + Fog) is a form of fog that occurs in areas, where the air contains a large amount of smoke.
- **Haze** is formed by water particles that have condensed in the atmosphere and visibility lie between 1 km to 2 km.
- **Frost** is the moisture on the ground surface that condenses directly into ice i.e. when condensation occurs below freezing point.

Clouds

- Clouds are a mass of small water droplets or tiny ice crystals.
- Clouds are classified according to their appearance, form and height. *There are four groups as follow*
 - High clouds** 6000 m to 12000 m.
 - Middle clouds** 2100 m to 6000 m.
 - Low clouds** below 2100 m.
 - Clouds with great vertical** extent 1500 m to 9000 m.

The different types of clouds are given Latin names, which are the combination of the following words :

- **Cirrus** means looking like a feather and used to describe the very high clouds.
- **Cumulus** means looking like a heap. It is used to describe the clouds having flat bases and rounded tops.
- **Stratus** means lying in level sheets. It is used for layer type of clouds.
- **Alto** means mid-level cloud.
- **Nimbus** means rain cloud.

High Clouds	Middle Clouds	Low Clouds	Clouds with Great Vertical Extent
Cirrus Composed of small ice crystal, white wispy and fibrous in appearance.	Alto Cumulus Composed of water droplets in layers and patches.	Strato Cumulus Large globular masses, bumpy looking, soft and grey in appearance forming a pronounced regular and sometimes wavy pattern.	Cumulus Round topped and flat based forming a whitish grey globular mass, consists of individual cloud units.
Cirro Cumulus Composed of ice crystals but rippled in appearance.	Alto Stratus Composed of water droplets, forming sheets of grey or watery looking clouds.	Nimbo Stratus Dark grey and rainy looking, dense and shapeless, often gives continuous rains.	Cumulo Nimbus They have a great vertical extent, white or black globular masses, whose rounded tops often spread out in the form of anvil. It is characterised by convectional rain, lightning and thunder.
Cirro Stratus Looks like a thin white almost transparent sheet, which causes the Sun and Moon to have halos.		Stratus These are low, grey and layered, almost fog like in appearance, bringing dull weather and often accompanied by drizzle.	

PRECIPITATION

On the basis of its origin, precipitation may be classified into three main types

- **Convectional Rainfall** It occurs due to thermal convection currents caused due to insolation heating of ground surface. e.g. Rainfall over Congo basin, Amazon basin and South-East Asia.
- **Orographic Rainfall** It occurs due to ascent of air forced by mountain barrier. e.g. Mahabaleshwar, Western Ghat receives about 600 cm rainfall.
- **Frontal Rainfall** It occurs due to upward movement of air caused by convergence of cold air masses against warm air masses. e.g. Heavy rainfall in North-West Europe due to convergence of cold continental and warm oceanic air.

WEATHER AND CLIMATE

- Weather refers to the sum total of the atmospheric conditions in terms of temperature, pressure, wind, moisture, cloudiness, precipitation and visibility of a particular place at any given time.
- The average weather conditions over a large area are called the Climate of a Place. The factors, which control the weather and the climatic conditions are latitude, altitude, unequal distribution of land and water, ocean current, air pressure and wind, mountain barrier, nature of ground surface, different types of atmospheric storms etc.

Name	Country/ Region
Sahara (Libyan, Nubian)	North Africa
Australian (Gibson, Simpson, Victorian Great Sandy)	Australia
Arabian (Rub-al-Khali, An-Nafud)	Arabia
Dast-e-Lut (Barren desert)	Iran
Dast-e-Kavir (Salt-e-Kavir) (Salt desert)	Iran
Desierto de Sechura	Peru
Atacama	North Chile
Patagonia	Argentina
Kalahari	Botswana
Namib	Namibia
Great Indian Desert	India

Isopleth	Reaction
Isohels	Sunshine
Isohyets	Rainfall
Isonif	Snow
Isocline	Slope
Isotherms	Temperature
Isobars	Equal pressure
Isobath	Equal depth in sea
Isohaline	Salinity
Isohypse (or contour lines)	Elevation above sea-level
Isodapane	Equal transportation cost
Isobronts	Thunderstorm at the same time

Grasslands	Countries	Grasslands	Countries
Steppe	Eurasia	Veld	South Africa
Puszta	Hungary	Downs	Australia
Prairie	USA	Canterbury	New Zealand
Pampas	Argentina		

Climatic Zone	Climatic Types	Rainfall	Natural Vegetation
Equatorial Zone (10°N-10°S)	Hot, wet equatorial	Rainfall all the year	Equatorial rain forest– special term selva is used to Ebony, mahogany, dye woods are found.
Hot Zone (30°N-30°S)	(i) Tropical Monsoon (ii) Tropical Marine (iii) Sudan type (iv) Desert (a) Saharan type (b) Mid latitude type	Heavy summer rain Summer rain Rain mainly in summer Little rain	Monsoonal forest– Teak sal, sisam, eucalyptus acacia savana (Tropical grassland)– grasses Desert vegetation and scrub– cacti
Warm Temperate Zone (30°N-45°S)	(i) Western margin (Mediterranean type) (ii) Central Continental type (Steppe type) (iii) Eastern Margin (a) China type (b) Gulf type (c) Natal type	Winter rain Light summer rain Heavier summer rain	Mediterranean forests– Oak, pine fir, eucalyptus, giant sequoia, cedar Steppe, temperate grass Warm, wet forests– Parapine, oak Parapine, oak Walnut, oak, hickory Parapine, eucalyptus, wattle
Cool Temperate Zone (45°N-65°S)	(i) Western Margin (British Type) (ii) Central Continental (Siberian Type)	Rain in autumn and winter Light summer rain	Deciduous forest–Oak, elm, birch, poplar, chestnut, maple Coniferous forest– Pine fir, spruce larch
	(i) Eastern Margin (Lauritian Type)	Moderate summer rain	Mixed forest– Oak, beech, maple
Cold Zone (65°N-90°S)	(ii) Arctic or Polar Mountain Climate	Very light summer rain Heavy rainfall (variable)	Tundra, mosses, lichens Alpine, fern, conifers

HYDROSPHERE

The water components of the Earth is called Hydrosphere. It includes the oceans, seas, lakes, ponds, rivers and streams. The hydrosphere covers about 70% of the surface of the Earth.

Storage Component	Total Percentage of Water
Oceans	97.6
Saline lakes and inland seas	0.008
Ice capes and glaciers	1.9
Ground water	0.5
Soil moisture	0.01
Lakes	0.009
Freshwater rivers	0.0001
Atmosphere	0.0009

Relief Features of the Oceans

Continental Shelf

- The shallow submerged extension of the continent is called the **continental shelf**. Extends to a depth of 100 fathoms (1 fathoms = 1.8 m). It has average width 70 km; **average slope** 17 ft/mile or about 1°.
- Continental shelf covers 7.5% area of the oceans. It extends over 13.3% of the Atlantic ocean, 5.7% of Pacific ocean and 4.2% of Indian ocean.
- In cases, where the mountains extend along the coast, the continental shelf will be narrower. About 20% petrol and gas are found here. They also provide the richest fishing ground in the world.

Continental Slope

- Extends seawards from the continental shelf. It has depth—200-2000 fathoms (3660m) and average slope 20 to 50 degrees.
- The boundary between shelf and slope is known as **Andesite Line**, on the names of the Andesite rock. Continental slope covers about 8.5% of the total ocean area.

Continental Rise

- Continental rise is an area at the foot of the slope, slightly rising due to the accumulation of debris transported over the slope.
- It has average slope— 0.5° to 1°. Oil deposits occur here.

Abyssal or the Deep Sea Plains

- It is the deepest and most extensive part of the oceanic floor having average depth of 3000m to 6000m. It covers about 75.9% of total oceanic area.

- Parts of the Abyssal plains are occupied by raised ridges or submarine mountains and by very deep trenches or canyons.

Deeps/Trenches

- Trenches are narrow and steep sides depressions. Trenches are formed, when two plates of the Earth's crust are moving together and one is being pushed down below the other.
- Marina trench (challenger deep) is the deepest trench in the world situated in the NW Pacific oceans, near philippines. It is approx 11 km deep.

Oceanic Ridges

- Oceanic ridges are of tectonic origin and they provide evidence in support of the Plate Tectonics Theory. They are formed by the volcanic activity along the spreading boundary of plates.
- These ridges are thousands of km long and hundreds of km wide mountain ranges on the oceanic floor.
- Their summits may rise above the sea level in the form of islands i.e. Iceland, Azores Island are such examples.

Coral Reefs

- Coral reefs are formed due to accumulation and the compaction of skeletons of lime secreting organisms known as **Coral Polyps**.
- Corals are found mainly in the tropical oceans and seas because they require high mean annual temperature of ranging between 20°C to 25°C.
- Corals do not live in deeper waters due to lack of sufficient sunlight and oxygen.

On the basis of nature, shape and mode of occurrence, the coral reefs are classified into three types :

- Fringing Reefs** It develop along the continental margins or along the islands.
 - The seaward slope is steep and vertical while the landward slope is gentle. These reefs run as a narrow belt of about 0.5 km to 2.5 km width.
 - They are usually attached to the coastal land, but occasionally they are separated from the shore by a shallow and narrow lagoon called 'Boat Channel'.
 - This type of reefs are found near Rameshwaram in the Gulf of Mannar.
- Barrier Reefs** Largest coral reefs off the coastal platform, but parallel to them.
 - The reef lies at a distance away from the coast. Hence, a broad lagoon develops between the reef and the shore. They are hundreds of kilometer long and few kilometers wide.
 - **The Great Barrier Reef** of Australia is the largest barrier reef in the world.